



10 CPU Cooler:

NZXT Kraken X60

It won't ever be acceptable to use anything other than liquid cooling to cool the Intel Core i7 4960X on our Ultimate PC and so we unleashed the Kraken on its arse, which makes sense owing to the fact that we had to cool a CPU that had a TDP of over 130 Watts. With a 280mm radiator, the Kraken x60 sat right inside the top mounting area of the case. A total of four silent fans were selected and mounted in a push-pull configuration to carry the hot air from the radiator. When we finally had our system up and running, we recorded an overall CPU idle temperature of 21°C at an ambient room temperature of 20°C which was nothing short of astounding for an all-in-one liquid cooler. The Kraken still held itself well when we stress-tested the CPU by running loops of Prime95 torture test and the 4960X hardly crossed a maximum load temperature of 52°C. Using the Kraken X60 meant that we could still have some headroom for future overclocking endeavours. Which you'll read about very soon. Online.

11 SSDs:

SanDisk Extreme II 240GB x 4

When someone drops four high-performance SSDs into your lap to do what you want with, you nonchalantly flip them aside and continue to play WoT. Once that person leaves however, you drop everything and slap them into the Ultimate Rig and set 'em all up in RAID 0 with all the excitement and energy of a kid discovering Half-Life 3 under his pillow in the morning. Owing to the limitations of the board, we had to set two on SATA3 and the other two on SATA2, but it couldn't be helped. These drives also served as our primary storage because, well, why the hell not? The Extreme II uses Marvell's - Monet 88SS9187 controller with 256MB DDR3-1600 DRAM onboard. We saw the SanDisk Extreme II RAID 0 SATA3 pair getting read and write sequential data transfer speeds of 907.46 MB/s and 822.70 MB/s respectively.



12 Case:

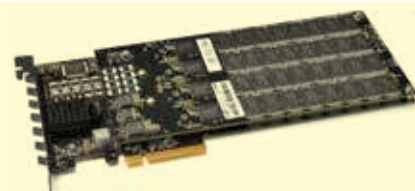
Corsair Obsidian 900D

An epic PC without an epic case is just a sad PC. The epic case for the Ultimate PC had to be the Obsidian 900D. A bottomless pit of a case, this one just swallows components whole and never spits them back out. We spent a night camped inside as we sat and assembled the components, setting up a hammock between the hard-drive bay and the back panel. With a case this large, assembly and cable management was never going to be an issue and it wasn't. If anything, some of the cables turned out to be too short to circumnavigate the girth of this case.

The 900D is capable of accommodating a 480mm radiator at the top, another at the bottom, a 240mm radiator at the side

and bottom again, a 140mm radiator at the rear and a 360mm radiator on the front. Whew! We can breathe again. We only needed a 280mm radiator however, and we mounted that on the top. The 900D can, of course, accommodate two PSUs, which was ideal. There was so much space left in the case even after we finished assembling the rig that it actually seemed empty, which is saying something. And yes, we put in 13 fans, which incidentally (but definitely not intentionally) is a nod to the 13 years that Digit's been alive and kicking. Could we have gone with another case? Yes. Would we? No.

The amount of room this case offered to play around was simply staggering. We would have loved to permanently setup residence in that case, but it was, unfortunately, too drafty for health. Too bad. We could have had so much fun.



13 PCIe SSD:

OCZ Z-Drive R4 CM88 1.6TB

After being blown away by the benchmark scores that we were seeing with this build, we decided that we weren't blown away enough and that an OCZ Z-Drive R4 that costs as much as a mid-size sedan was a more sensible option.

There are few instances in a mortal's life when one sees something as powerful as the OCZ Z-Drive R4. Think of this drive as the Pagani Zonda of the computer world, which is also proportionately as expensive and as powerful as a high-end PC in itself. It costs just under ₹9,90,000 and no, it isn't diamond studded or gold plated but is in fact, coated with something more precious, 8 x SF-2282 (SandForce) controllers and it also uses two PCBs to accommodate all those controllers. With so many controllers available, the drive has a total of 128 Intel 25nm MLC NAND devices on it. Since we don't want to bore you with more specifications, take a brief look at its performance. We assure you that you'll never see such scores again. Not for a long time yet.

Exhibit A: Crystal Disk Mark OxFill Sequential Read speed: 1252 MB/s; Sequential Write Speed: 1731 MB/s.
Exhibit B: AS SSD overall benchmark score: 3530; Sequential Read speed: 2419.69 MB/s with an access time of 0.063 ms; Sequential Write speed of 1847.16 MB/s with an access time of 0.161 ms.

The results were synonymous with real world tests too. We recorded read and write speeds of 962 MB/s and 1410 MB/s respectively for a sequential file transfer of 8GB (that's 8 seconds for an 8GB file by the way). Assorted file read speeds were also equally crazy. We didn't have any relative performance numbers to quantify the result because we never tested something this fast. This is what happens when you pair an enterprise grade component with a consumer grade ultra high-end PC.

On a more practical note, you really don't need this SSD for any possible home usage scenario. But who cares? This is the Ultimate PC and it needs the ultimate components. Nothing more, nothing less.

Note:

We actually made multiple builds of the Ultimate PC as we could only run a maximum of four full sized PCIe cards on the Rampage IV Black Edition. We used the OCZ Z-Drive R4 CM88 along with a tri-SLI GTX 780Ti setup for one build and 4 x SanDisk Extreme II along with a quad-SLI 780Ti setup for another.